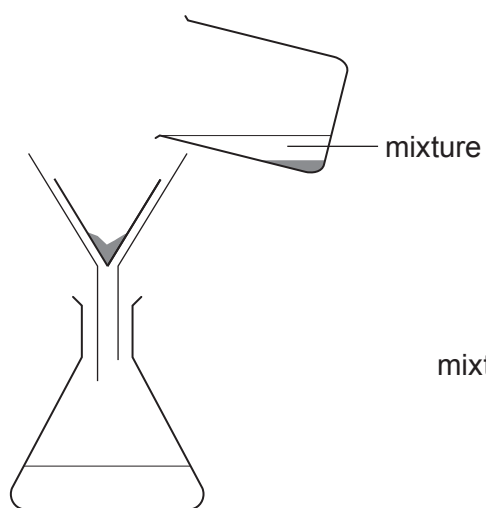
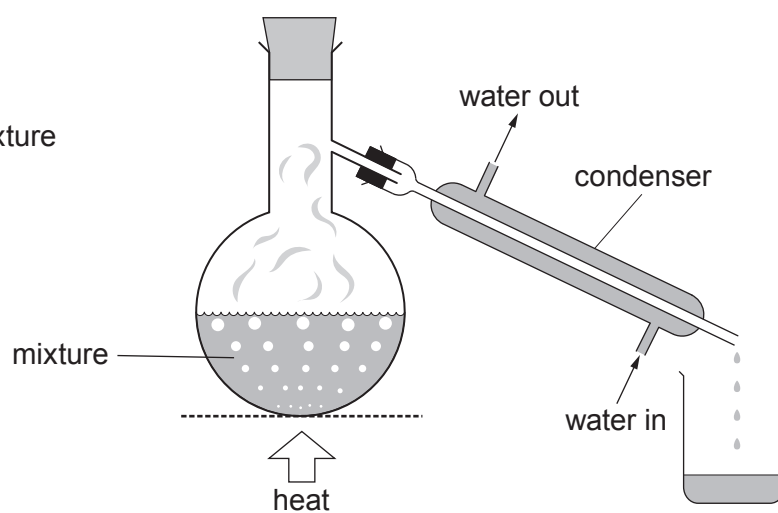


Answer **all** questions.

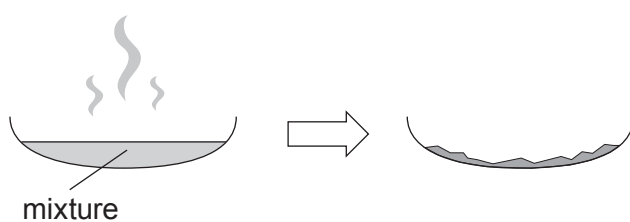
1. (a) The diagrams show four methods, **A**, **B**, **C** and **D**, used to separate different mixtures.



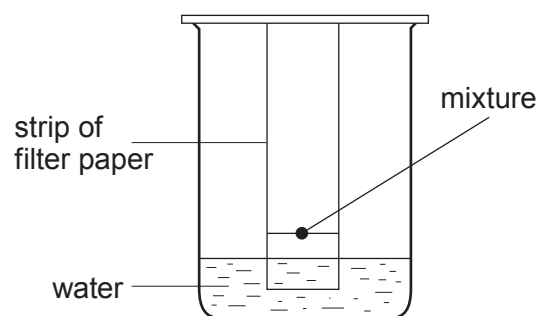
Method **A**



Method **B**



Method **C**



Method **D**



- (i) Choose from the box the names of methods **B** and **D**.

[2]

distillation

chromatography

filtration

evaporation

boiling

Method **B**

Method **D**

- (ii) Give the letter, **A**, **B**, **C** or **D**, of the method used to

[3]

remove sand from water

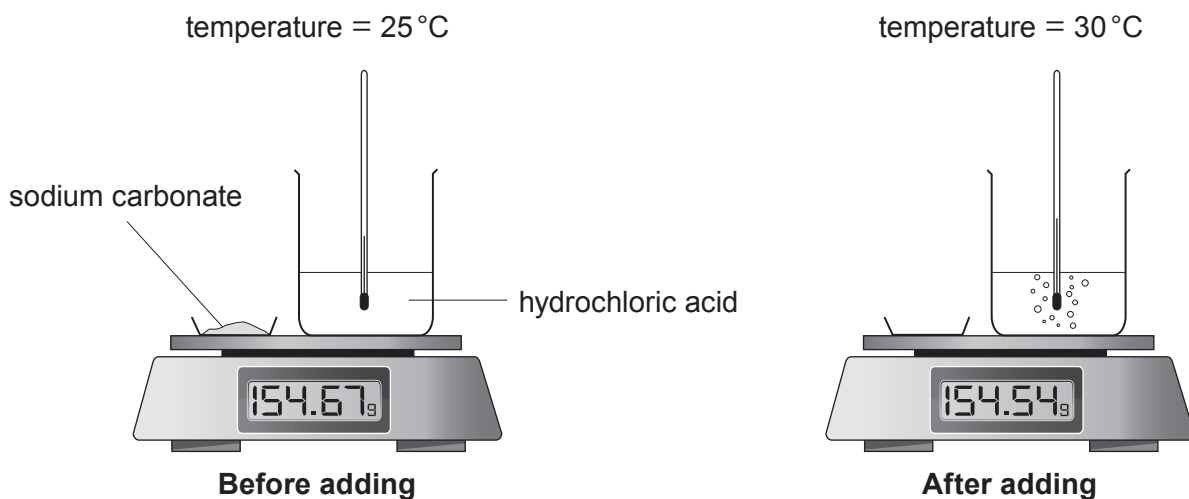
obtain pure water from sea water

separate red and yellow dyes



- (b) Sodium carbonate reacts with dilute hydrochloric acid forming sodium chloride, water and carbon dioxide.

The diagrams show the apparatus before and after sodium carbonate is added to hydrochloric acid.



Tick (✓) **two** observations that show a chemical reaction is taking place.

[2]

The solid stays the same

☐

A gas is formed

☐

A temperature change occurs

☐

The mass of the beaker and contents stays the same

☐